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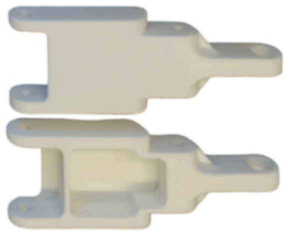
Hands-on Tutorial: 3D Tactile Sensor Integration in Robotic Fingers for Smart Manipulation

1. 3D Printed Parts list

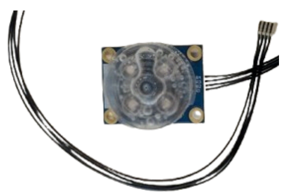
Part	Quantity	Image
Internal Rod	2	
External Rod	2	
Distal Rod	2	
Removable Tip*	2	
Distal Shell*	2	

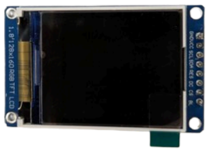
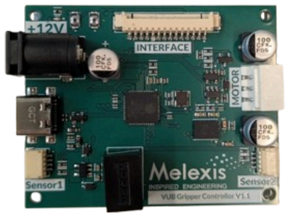
Top Plate	1	
Motor Flange	1	
Cam	1	
Driving Rod	2	
Bottom Plate	1	
Head Housing*	1	

Interface panel for Head Housing*	1	
Top panel for Head Housing*	1	
Potentiometer Knob*	1	

**These parts have been added and are not from Pollen Robotics.*

2. Components list

Component	Quantity	Image
Feetech STS3215 Servo Motor, JST Wire and screws	1	
Touch Sensor, cables and cover of the sensor	2	

Interface PCB and his cable	1	
LCD Screen	1	
Joystick	1	
Potentiometer	1	
Controller PCB	1	
Screw M2 x 8	16	
Hex Nut M2	4	
Spacer	1	

Thermoplastic Screw 2.5 × 8	24	
Screw M3 × 0.50 × 10	4	
Parallel Pin 3 × 16 h6	4	
Parallel Pin 3 × 24 h6	4	
Insert M2	8	
Insert M3	4	
Plain Bearing 3 x 6 x 5	4	
Tools : - Screwdriver KL-1 - Hex Wrench 2.5mm	2	 

Test objects	3	
Power supply 12V, 5A	1	
USB-C cable	1	

3. Pre-Assembly

Insert the bearings in the provided bores

The bearing is supposed to fit by forcing with hand, or pressed onto a solid surface. No tool is needed here.

If it’s too hard, file or drill the hole a bit and repeat so that it fits tighten.

If the hole is too wide, put a drop of glue

M3 Brass inserts

Insert **M3 brass inserts** using a soldering iron.

Make sure that you are on the right side.

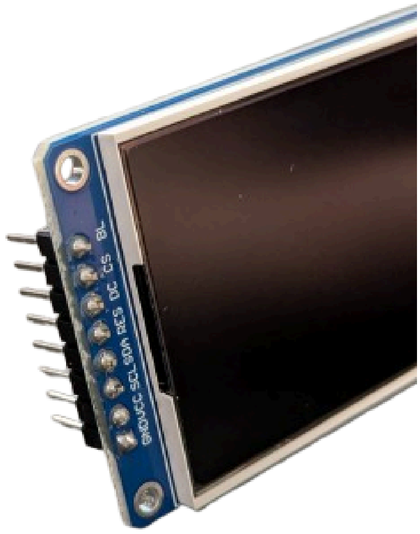
M2 Brass inserts

Insert **M2 brass inserts** using a soldering iron.

Screen Soldering

Position the black connective as shown in the picture.

Solder each pin to the card, on the screen side.



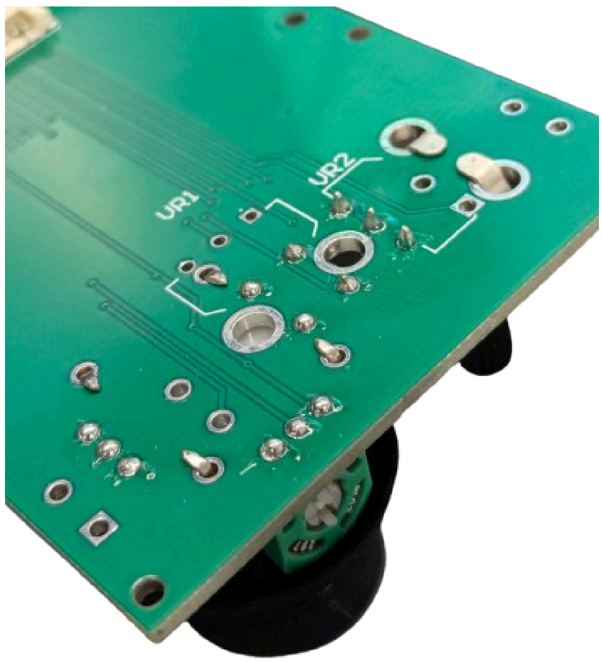
Interface PCB Soldering

Position the female black connective on the Interface PCB, as shown in the first picture.

Solder each pin on the back of the PCB.

Position the Joystick on the PCB.
Bend the four metallic stabilizers and solder the pins to the PCB.

Position the Potentiometer to the PCB.
Bend the two metallic stabilizers and solder the pins to the PCB.



4. Assembly Guide

Follow the steps below to correctly assemble the Parallel-Finger Gripper.

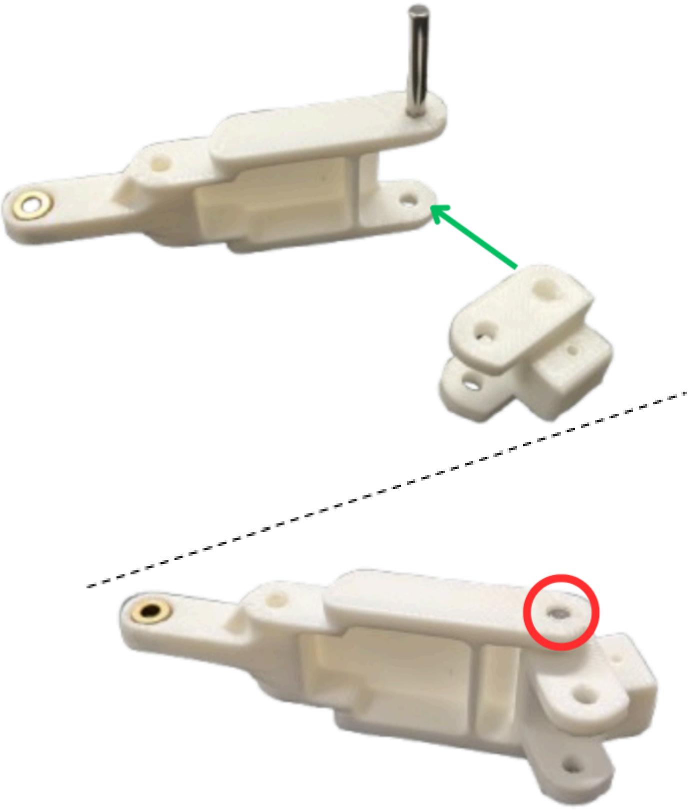
Step 1: Finger Assembly

Note: Repeat this entire process for the other phalanx to build both fingers.

First Phalanx Construction

Take the **Internal Rod (bag A)** and position the **Distal Rod (bag A)** as shown in the picture (green arrow).

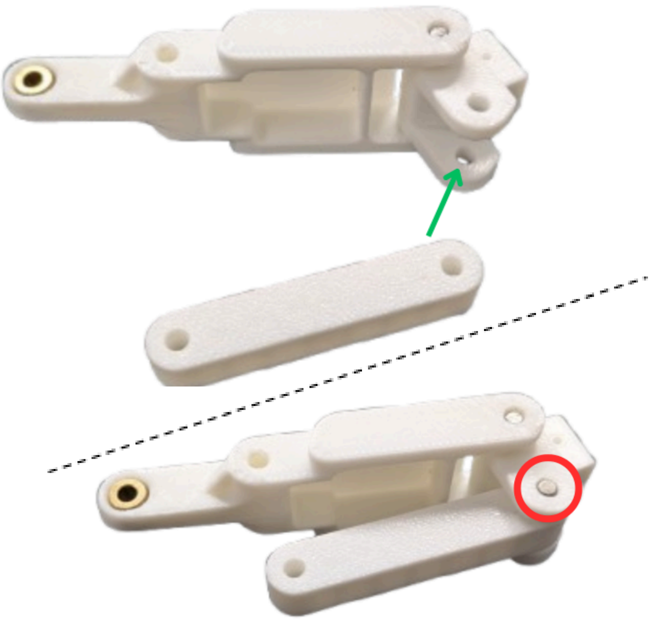
Insert a **Parallel Pin 24mm (bag L)** into the hole (red circle).



Position the **External Rod (bag A)** as shown in the picture (green arrow).

Ensure the External Rod is placed on the hollow side of the phalanx.

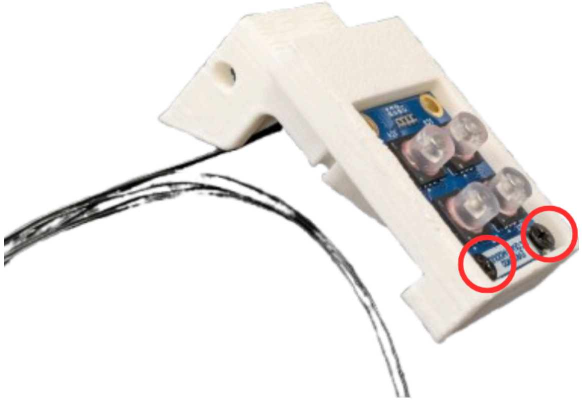
Insert the **Parallel Pins 16 mm (bag L)** into the hole (red circle).



Position the **Touch Sensor (bag G)** onto the **Removable Tip (bag B)**.

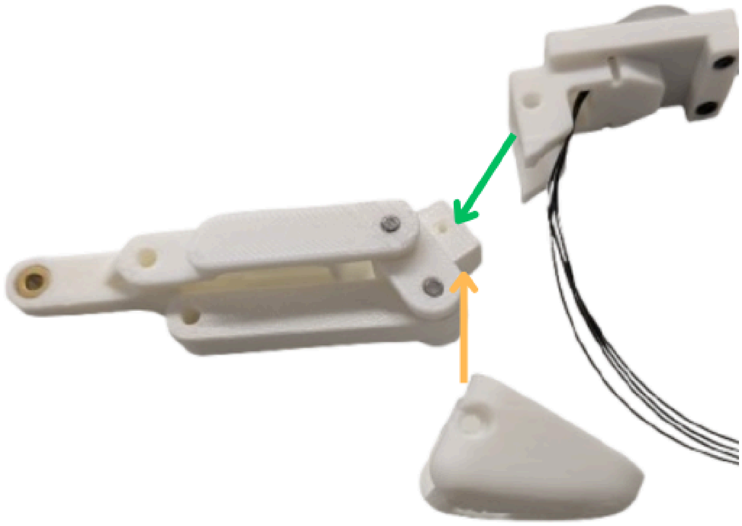
Route the cable through the hole located near the sensor slot.

Secure the sensor using two **M2 x 8 mm screws** and their corresponding **nuts (bag J)** (red circles). No need to screw too hard.



Slide the **Tip Support (bag B)** onto the dovetail joint (green arrow).

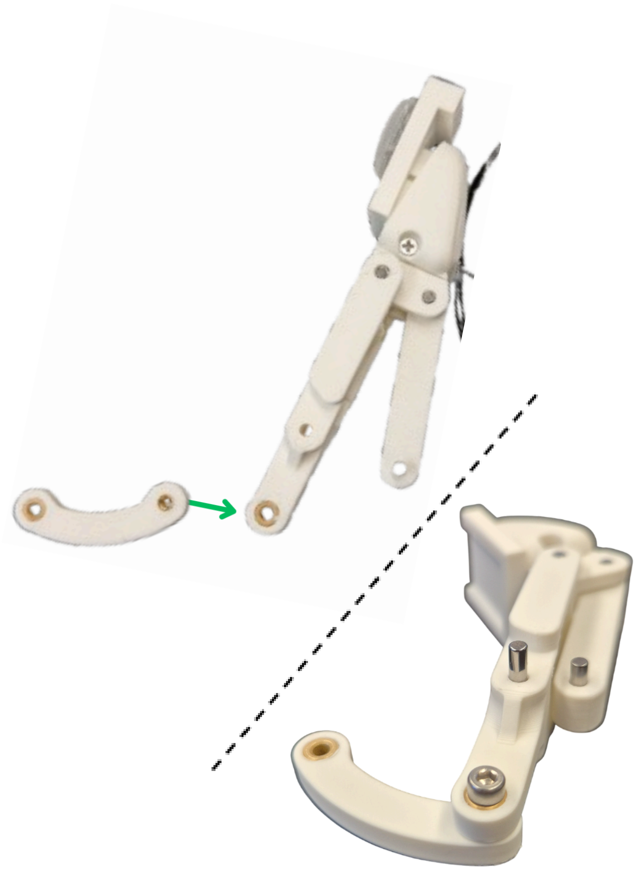
Route the cable through the **Distal Shell (bag B)** and place it over the assembly (orange arrow)/ Finally, secure it by tightening 2 **Thermoplastic Screws 2.5 x 8mm (bag K)**.



Attaching the Driving Rods

Right Finger: Place the **Driving Rod (bag D)** as shown in the picture (green arrow).

Secure it using a **Screw M3 x 10mm (bag K)** with the allen key.



Attaching the Driving Rods

Left Finger: Place the **Driving Rod (bag D)** as shown in the picture.
Be careful here, the orientation of the Driving Rod is different from the Right Finger.

Secure it using a **Screw M3 x 10mm (bag K)** with the allen key.

Both Fingers

Verification: Your assembled left and right fingers must look like the provided picture.

<u>Step 2:</u> Motor configuration and Installation

First configure the motor.

To do so :

1. Take the Interface PCB **PCB interface** **(bag H)** and connect it to the **Controller PCB** **(bag I)**.

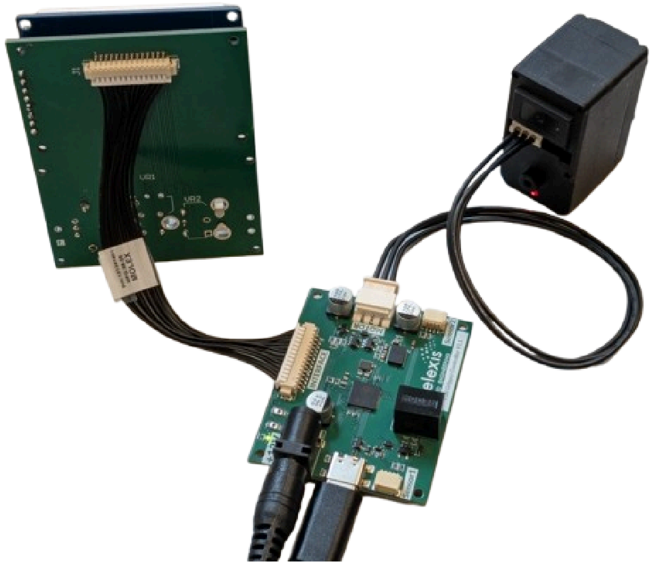
2. Place the **LCD screen**, the **Joystick Hat** **(bag H)** and the **Potentiometer Knob** **(bag F)** on the Interface PCB.

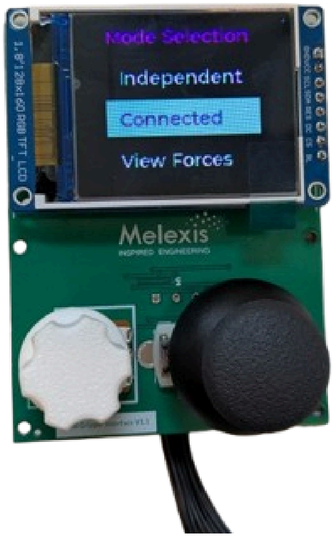
3. You can now connect the **Servo Motor** to the Controller PCB using his **cable** (in the Feetech box).
   Connect the usb-c and the power supply to the PCB.

4. Follow the steps shown on the screen.

5. Once you are in the menu interface, select “Connected”.
   You should see a message that says that you can launch the Python Code.

6. On your computer :
   The provided script is in
   Folder/Codes/python_codes





Use the following command to run it :

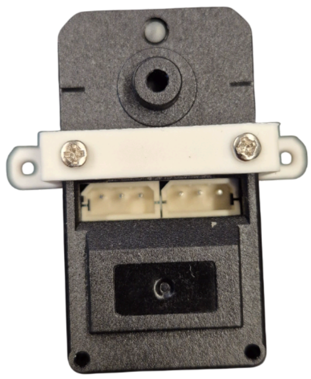
python3 Motor_initialisation.py

You should see the motor turn and go back to its initial state. The motor is now initialized in an “open finger” position.

Disconnect all the cables by pulling on the connecters (not the cables), and put away the different parts in their respective bags.
We’ll use them again later.

Motor Flange

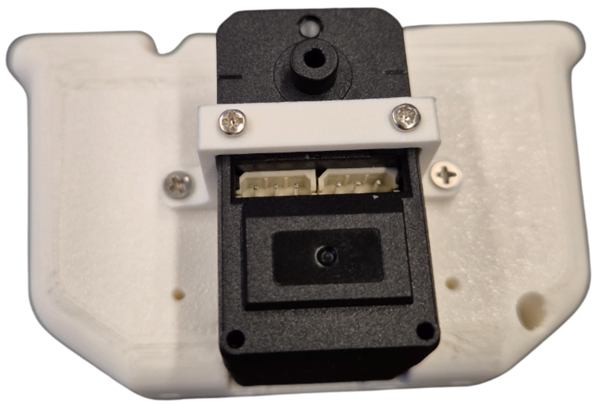
Secure the **Motor Flange (bag C)** to the **Servo Motor** using the self-tapping screws provided by Feetech (in the Feetech Servo Motor box).

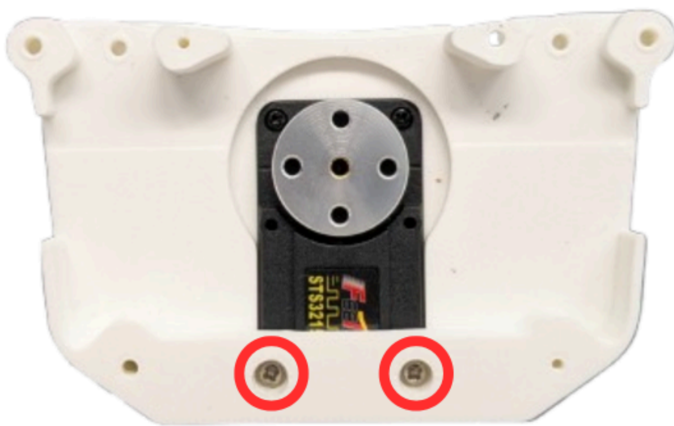


Motor on Palm: Insert the motor into its designated slot on the palm of the **Top Plate (bag C)**.

Secure it from the top using 2 **Thermoplastic Screws 2.5 x 8mm (bag K)**.

Flip the palm over and insert 2 additional **Feetech self-tapping screws** from the bottom, as indicated in the picture (red circles).



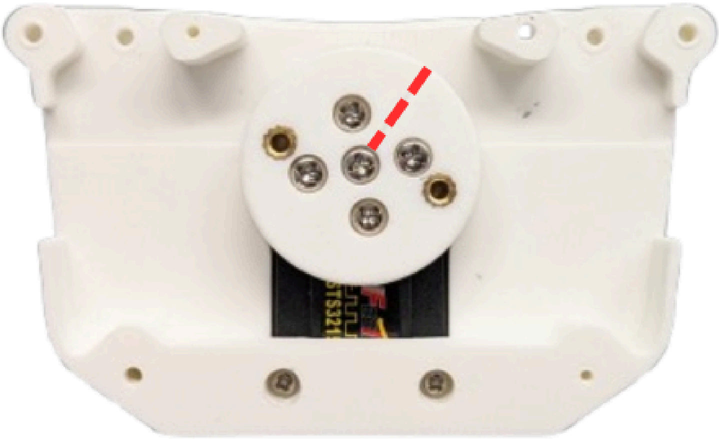


<u>Step 3:</u> Cam Assembly

Place the **Cam** **(bag D)** onto the motor.

Ensure the alignment mark is pointing as shown in the pictures (red dot line).

Secure the Cam using 5 **Feetech screws** (in the Feetech Servo Motor box).



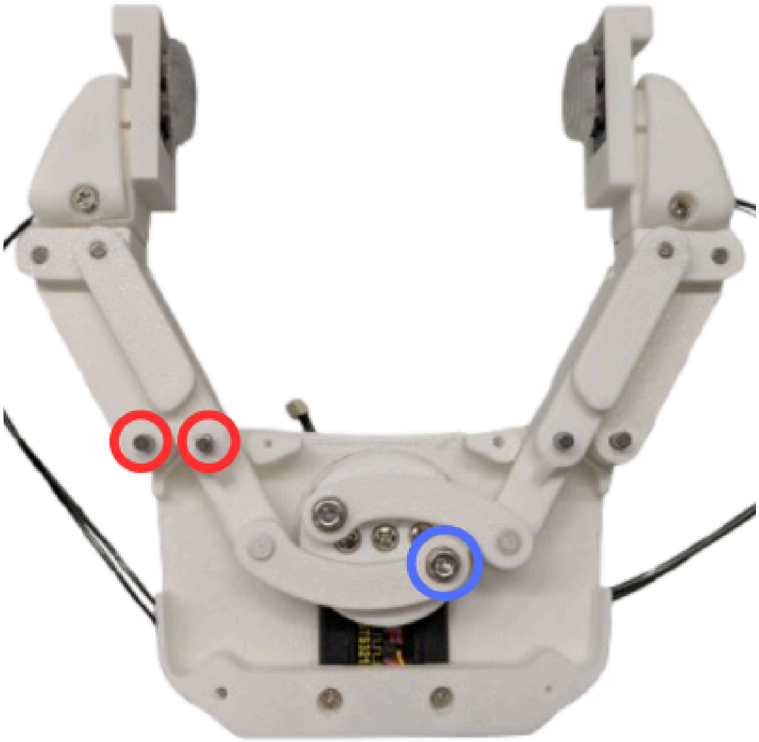
<u>Step 4:</u> Finger Integration

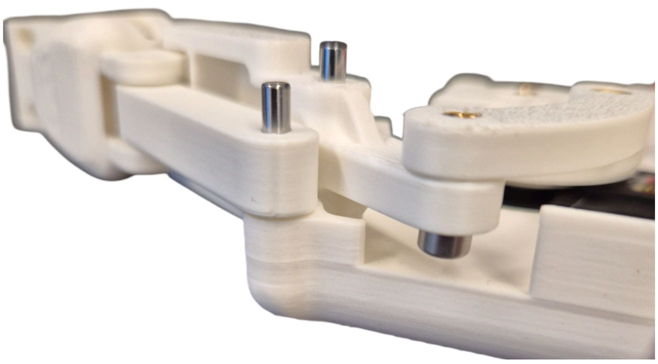
Flip the main assembly over.

Note: Because the assembly is flipped, the right finger is now located on your left.

Insert the Parallel Pins into the indicated holes into the finger (red circles).

Attach the **Driving Rod** **(bag D)** of each finger to the **Cam** (blue circle) and tighten them using a **Screw M3 x 10mm** **(bag K)** for each.





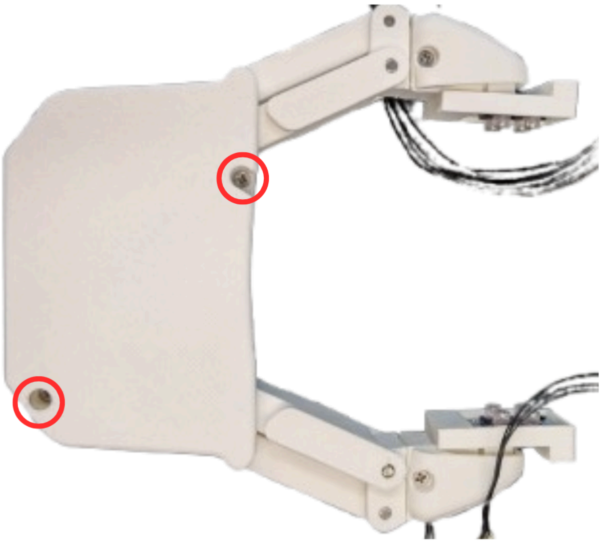
Verification: Your fully attached left and right fingers must look like the provided picture.

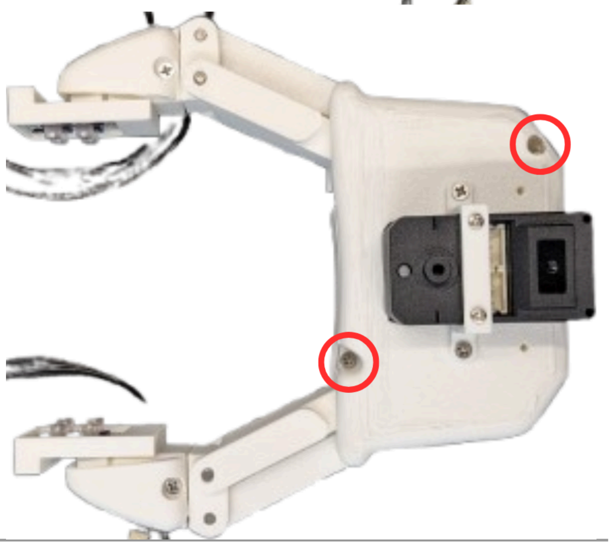
<u>Step 5:</u> Shell Closing

Place the **Bottom Plate** **(bag E)** to close the palm assembly. Make sure that the two plates are firmly pressed together.

Secure the Bottom Plate using 4 **Thermoplastic Screws 2.5 x 8mm (bag K).**

Place two other screws on the front and two on the opposite side (red circles).





<u>Step 6:</u> Head Housing and PCBs

Attach the Head Housing support to the newly assembled hand.

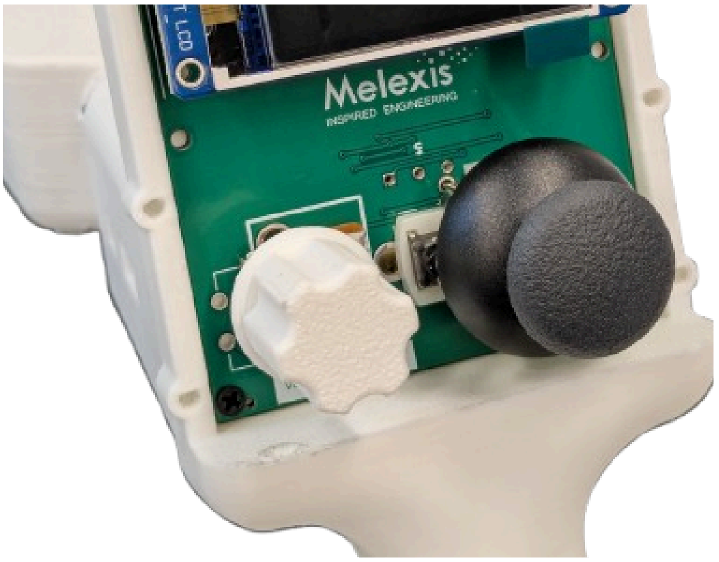
To do so, use five **Thermoplastic Screw 2.5 × 8mm (bag K)** and insert the screwdriver through the opening visible in the picture.

Position the **PCB interface (bag H)** and run the cable through the slot on the back as shown in the image.

Secure the PCB using three **Screw M2 x 8 mm (bag J)** in the corners and one **Spacer (bag J)** in the upper-left corner.

Insert the **LCD screen (bag H)** onto its pins and secure it using one **Screw M2 x 8 mm (bag J)** in the upper-left corner.

You can then place the **Knob (bag F)** onto the potentiometer and the **Joystick Hat (bag H)** if it's not already done.



Install the **Interface Panel (bag F)** and secure it with four **Thermoplastic Screw 2.5 × 8mm (bag K)**.

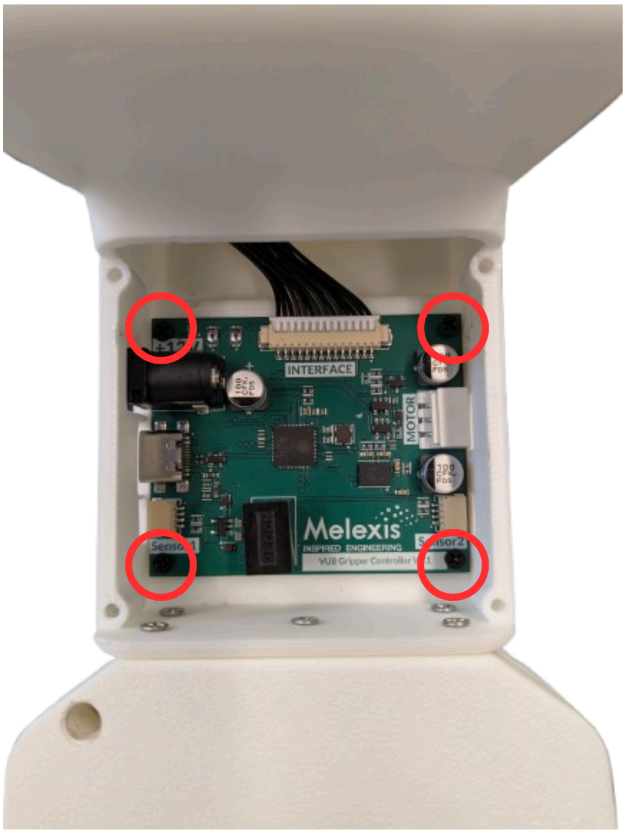
Insert the **Controller PCB (bag I)** into this slot.

To do so, slightly tilt the PCB so the power supply connector enters first.

Connect the black cable from the Interface PCB.

Then, secure the PCB using four **Screw M2 x 8 mm (bag J)**.

There is no need to overtighten, the PCB only needs to remain stable.



Install the **Top Panel (bag F)** using four **Thermoplastic Screw 2.5 × 8mm (bag K)**.

Step 7: cables

You can now connect the cables to the second PCB. Check the right orientation of the cable before connecting it.

For the left side (picture 1):

- Left Finger Touch Sensor, you can use the hex key to help with the connection
- Motor

For the right side (picture 2):

- Power Supply
- USB-C Cable
- Right Finger Touch Sensor

Everything is ready, you can now use your robotized hand using 3D touch sensors!

Enjoy!